Electric Potential

Friday, March 15, 2024 2:49 PM

I

$\vec{F} = k\frac{q_1 q_2}{r^2}\hat{r}$; E_2 F q_1 q_2 F E_1

$\vec{E} = \frac{\vec{F}}{q}$ $\vec{E} = k\frac{q}{r^2}\hat{r}$ (point charge)

geometrical objects

ROD ; RING etc.

$\int k\frac{dq}{r^2}\hat{r} = \vec{E}$

charge densi

1D: λ $\left\{\frac{Q}{L}\right\}$

2D: σ $\left\{\frac{Q}{A}\right\}$

use

II

Gauss Law $\oint \vec{E}\cdot d\vec{A} = \frac{q_{in}}{\varepsilon_0}$ → works best when w

to calc. $\vec{E}$

few cases → point ; sphere

∞ long ROD/CYLINDER

∞ large surface

"smart" gauss → symetrical

Electric Potential (Potential) V (volt)

Phys I

g, m, h, g, g

Work done $= mgh = \boxed{W = -\Delta U}$

$W = F \cdot s$

$mg = F$

$g = \frac{F}{m}$

$U = mgy$ (pot

Phys 2

E, +q, d, +q

$W = \vec{F}\cdot\vec{s} = qEd = -\Delta U$

$E = \frac{F}{q}$

$*\{\, qV = U$

charge, elec. potential = pot. energy

$qEd = -\Delta U = -\Delta(qV) = -q\Delta V \Rightarrow$ Ed

* $qV = U$

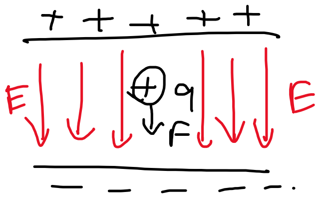
$[C]\ V = [J] \Rightarrow V = \left[\frac{J}{C}\right]$

electric potential

$\left[\frac{N}{C}\, m\right] =$

$W = -\Delta U$ (true for any conservative force)

↳ grav. force /



E = uniform / const. between parallel pl

$Ed = -\Delta V$ (when E is con

when $E \neq$ const

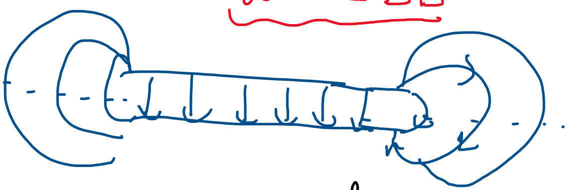
$$\boxed{-\Delta V = \int \vec{E}\cdot d\vec{r}}$$

$W = \vec{F}\cdot\vec{s} = \int$

$-\Delta U = W = q\vec{E}\cdot\vec{s} = q$

↳ $-\cancel{q}\Delta V = \cancel{q}\int \vec{E}\cdot$

$W = -\Delta U$

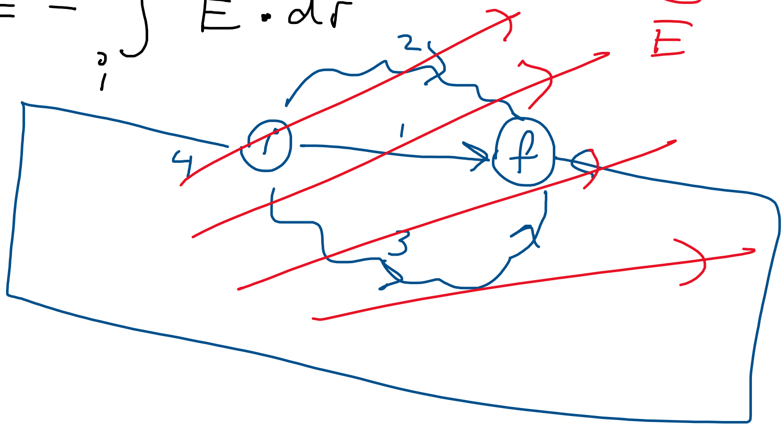


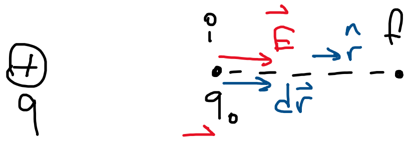
U = Energy = Joule =

1 eV =

$$\Delta V = -\int_i^f \vec{E}\cdot d\vec{r}$$

$V_f - V_i$





$$\vec{E} = \frac{kq}{r^2}\hat{r}$$

$$\Delta V = -\int_i^f \vec{E}\cdot d\vec{r} = \int_i^f E\, dr\, c$$

$$= -\int \frac{kq}{r^2}\hat{r}\cdot dr\,\hat{r}$$

1

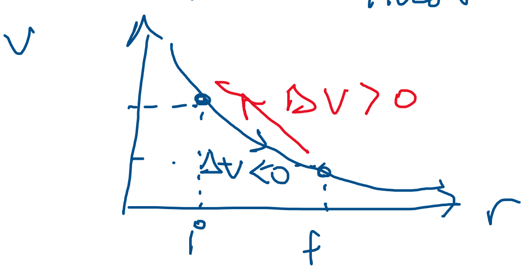
$$\Delta V = -\int \frac{kq}{r^2}\,dr \qquad -\int \frac{kq}{r^2}\,dr\;\underbrace{\hat{r}\cdot\hat{r}}_{1}$$

$$= -kq\int_{r_i}^{r_f}\frac{dr}{r^2} = -kq\left[-\frac{1}{r}\right]_{r_i}^{r_f} = kq\left[\frac{1}{r}\right]$$

$$\Delta V = \frac{kq}{r_f} - \frac{kq}{r_i}$$

$$V_f - V_i = \frac{kq}{r_f} - \frac{kq}{r_i} \quad \Rightarrow \quad V_f = \frac{kq}{r_f}, \quad V_i = \frac{kq}{r_i}$$

$$V = \frac{kq}{r} = \frac{q}{4\pi\varepsilon_0 r}$$ point charge



$\oplus$

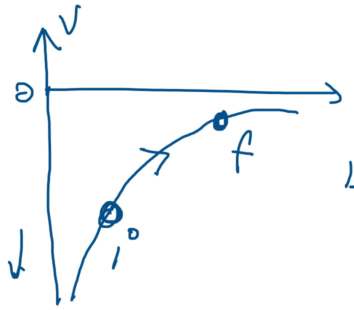
f ← i ⇒ Δ

i → f

$\Delta V < 0$

$\oplus \longrightarrow$ r $\qquad V = \frac{kq}{r} \qquad V \to 0 \quad r \to \infty$

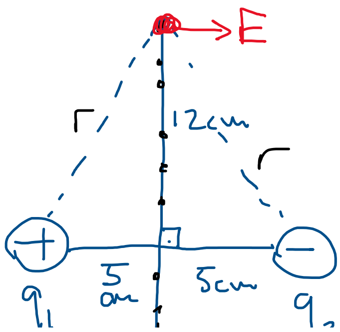
$\ominus \longrightarrow \qquad V = -\frac{kq}{r}$



$\Delta V > 0$

$$-\int \vec{E}\cdot d\vec{r} = \Delta V$$

$\hookrightarrow \frac{kq}{r^2}$ or $\frac{k(-q)}{r^2}$



$q_1 = +12\,nC$

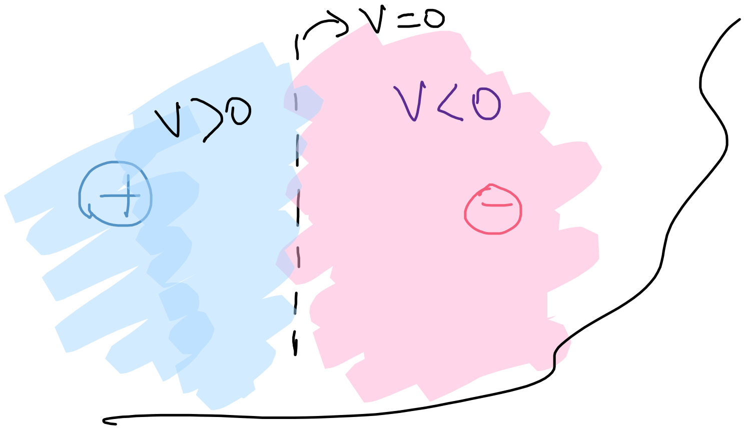
$q_2 = -12\,nC$

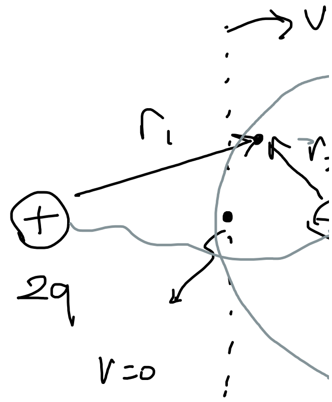
$V_0 = ?$ $\boxed{V_0 = V_1 + V_2}$

$r = 13\,cm$

$$V_0 = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} = \frac{k\,(12\,nC)}{0.13}$$

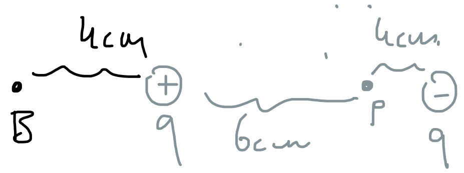
$\to V=0$





Ex. $q_1 = 2q$

$q_2 = -q$



$$V_P = \frac{kq_1}{r_1} + \frac{kq_2}{r_2}$$

$q = +12\,nC$ $\to 10^{-9}$ $\quad q = -12\,nC$

$$= kq\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

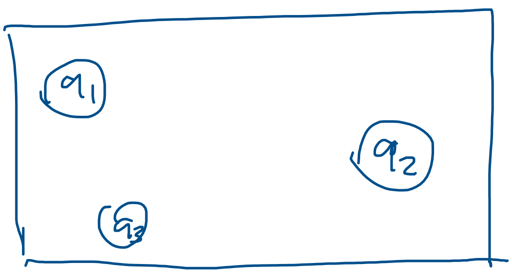
$$V_P = 9\times10^{9} \times 12\times10^{-9}\left(\frac{1}{0.06} - \frac{1}{0.04}\right) = -900\,V$$

$$V_B = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} = 9\times10^{9}\times12\times10^{-9}\left(\frac{1}{0.04} - \frac{1}{0.14}\right) = 193\ldots$$

Equipotential surfaces are surfaces where total

$U = qV$ $\Rightarrow$

pot. energy

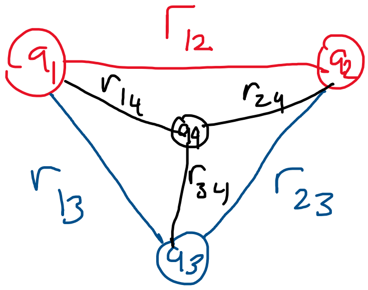


system

$q_1 \xleftrightarrow{r} q_2$ } system $\quad U = q_2 V_1 = q_2 \frac{kq_1}{r} = k\ldots$

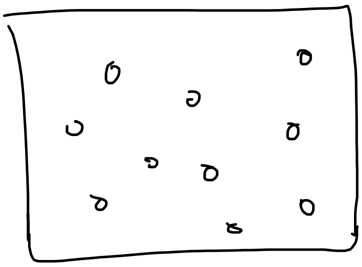
$U = q_1 V_2 = q_1 \frac{kq_1}{} = k\ldots$

for 3 point charges



$$U = k\frac{q_1 q_2}{r_{12}} + k\frac{q_2 q_3}{r_{23}} + k\frac{q_1 \ldots}{\ldots}$$

$$+ k\frac{q_4 q_1}{r_{14}} + k\frac{q_4 \ldots}{r \ldots}$$



$$U = \sum_{i=1}^{N} \sum_{j>i}^{N} k \frac{q_i q_j}{r_{ij}}$$

$q = 10\,nC$

q - - r - - · $1V$

$$V = \frac{kq}{r}$$

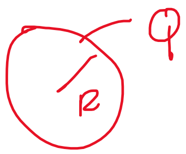
ⓔ - - - - · $1V$

$$\Delta V = -\int_i^f \vec{E}\cdot d\vec{r}$$

charged metal sphere

$V(r<R) = ?$

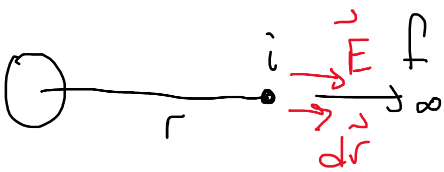
$V(r>R) = ?$



Reminder

$E = 0 \quad (r<R)$

$E = \frac{kq}{r^2} \quad (r>R)$

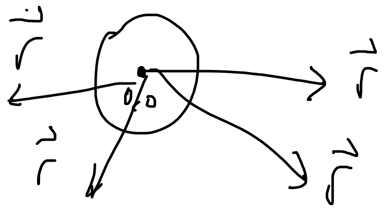


$$-\int_r^\infty \vec{E}\cdot d\vec{r} = \Delta V = -\int \frac{kq}{r^2}\,\hat{r}\cdot \ldots$$

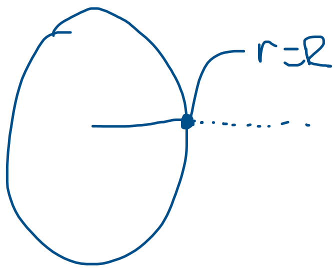
$$\Delta V = -kq\left[-\frac{1}{r}\right]_r^\infty = kq\left[\underbrace{\frac{1}{\infty}}_{\to 0} - \frac{1}{r}\right] = \Delta V = V_f - \ldots$$

$$V(r) = \frac{kq}{r} \quad (r>R) \quad \swarrow \quad kq\left[0 - \frac{1}{r}\right] = 0 - V(r \ldots$$

outside



$$V(r) = \frac{kq}{r} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$



$$V(r=R) = \frac{kq}{R} \Big\} = \text{non zero}$$

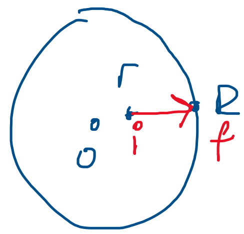
at r=R

$V(r<R)$

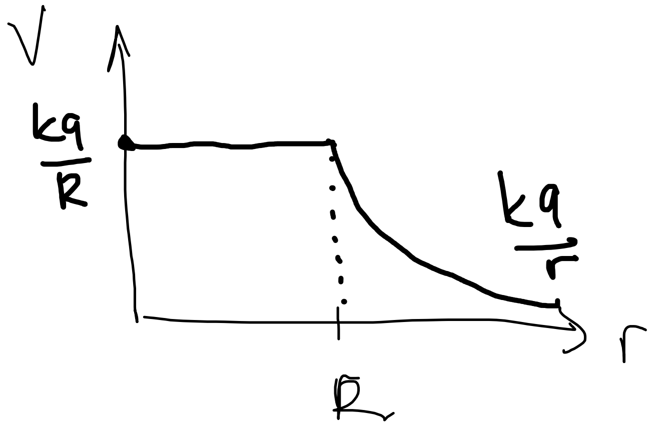
$\Delta V = -\int$

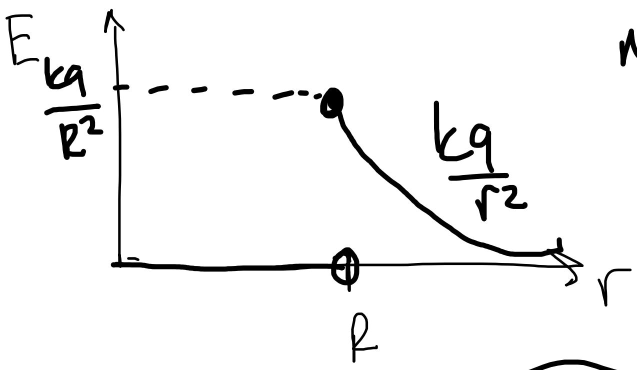
know E

$\Delta V = -$



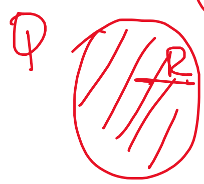
$V(R) = \frac{kq}{R} =$





metal, charged sphere

uniformly charged dielectric (insulator) sphere.



$V(r<R) = ?$

$V(r>R) = ?$

$$\Delta V = -\int \vec{E}\cdot d\vec{r}$$

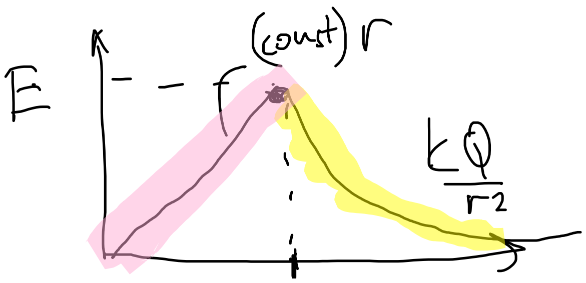
$$\rho = \frac{Q}{V} = \frac{Q}{\frac{4}{3}\pi R^3}$$

$\rho = ch$

kn

E

E



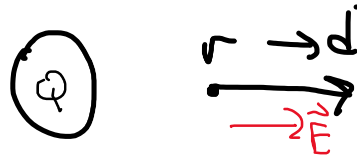
E Q $\frac{1}{3}$ r ρ

R

$E = \frac{\dots}{\underbrace{\frac{4\pi\varepsilon_0}{3}}\, R^3} = \frac{\dots}{3\varepsilon_0}$

$V(r>R) \Rightarrow$ same as metal sphere

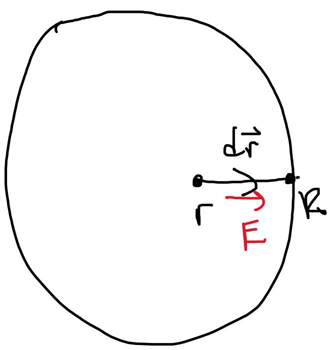
$$\Delta V = -\int \vec{E}\cdot d\vec{r} = -\int_r^{\infty} \frac{kQ}{r^2}\,\underbrace{\hat{r}\cdot d\vec{r}}_{dr}$$



$$\underbrace{V_{\infty} - V(r) = -kQ\left[-\frac{1}{r}\right]_r^{\infty}}\; \Rightarrow \quad V(r) = \frac{kQ}{r} \quad (r>R) = \text{sa}\dots \rightarrow r = \dots$$

$V(r<R)$ inside

$$\vec{E} = \frac{\rho}{3\varepsilon_0}\, r\,\hat{r} = \frac{\rho\,\vec{r}}{3\varepsilon_0}$$



$$\Delta V = -\int_r^R \frac{\rho}{3\varepsilon_0}\, r\,\underbrace{\hat{r}\cdot d\vec{r}}_{dr} = -\int \frac{\rho}{3\varepsilon_0}\dots$$

$$\Delta V = -\frac{\rho}{3\varepsilon_0}\int_r^R r\,dr = -\frac{\rho}{3\varepsilon_0}\left[\frac{r^2}{2}\dots\right.$$

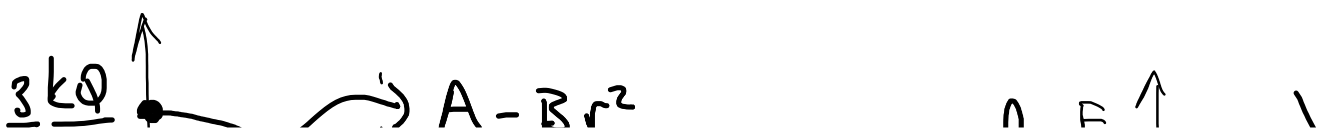
$$\underbrace{V(R) - V(r) = -\frac{\rho}{3\varepsilon_0}\left[\dots\right.}$$

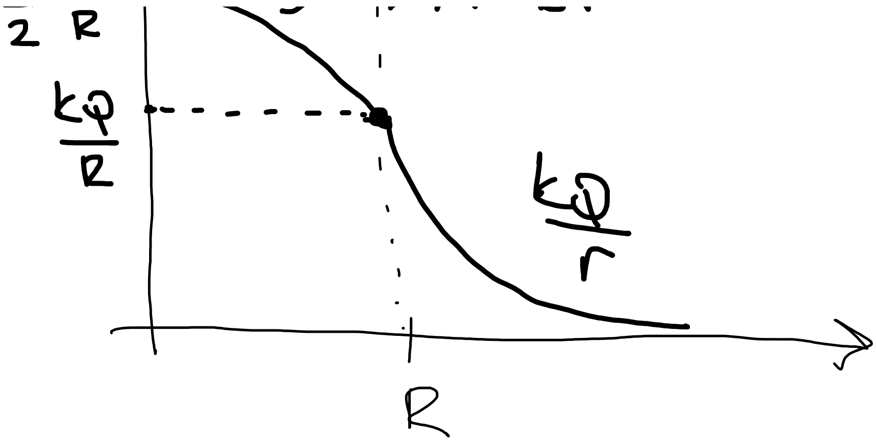
$$V(r) = V(R) + \frac{\rho}{3\varepsilon_0}\frac{1}{2}\left[R^2 - r^2\right]$$

$$V(R) = \frac{kQ}{R} = \frac{Q}{4\pi R\varepsilon_0} = \frac{Q}{\underbrace{\frac{4}{3}\pi R^3}_{\rho}}\,\frac{R^2}{3\varepsilon_0} = \frac{\rho R^2}{3\varepsilon_0}$$

$$V(r) = \frac{\rho R^2}{3\varepsilon_0} + \frac{\rho R^2}{6\varepsilon_0} - \frac{\rho r^2}{6\varepsilon_0} = \frac{3\rho R^2}{6\varepsilon_0} - \frac{\rho r^2}{6\varepsilon_0} = \dots$$

$$V(r=0) = \frac{3\rho R^2}{6\varepsilon_0} = \frac{\rho R^2}{2\varepsilon_0} = \frac{3Q}{8\pi\varepsilon_0 R} = (1.5)\,\frac{Q}{4\pi\varepsilon_0 R} = \frac{3}{2}\dots$$





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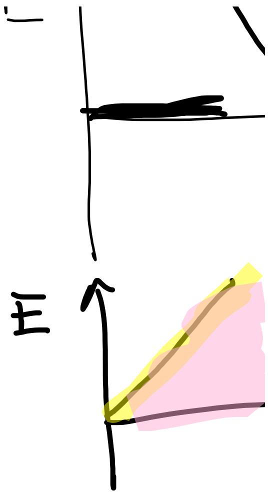
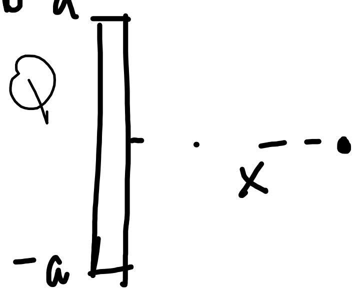


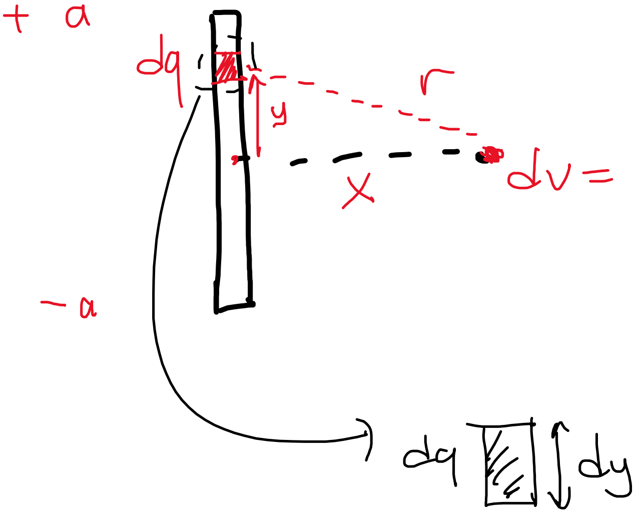
RoD a



$$E = \int \frac{k\,dq}{r^2}$$

dq — r — $dV =$

$$V = \int dV = \int \frac{k\,dq}{r} \Rightarrow$$ witho ve



$$dV = \frac{k\,dq}{r} = \frac{k\,dq}{\sqrt{x^2+y^2}}$$

$$\int \frac{k\,dq}{r} ?$$

$$\lambda = (1D) =$$

$$dq = \lambda\,dy$$

$$dq = \frac{Q}{2a}\,d$$

$$\int \frac{k\,dq}{r} = \int k\,\frac{Q}{2a}\,dy\,\frac{1}{(x^2+y^2)^{1/2}} \rightarrow \frac{kQ}{2a}\int_{-a}^{+a}$$

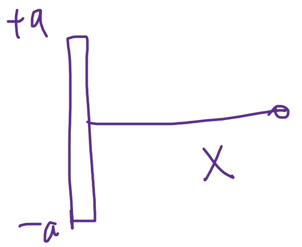
inte

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln\left(x + \sqrt{x^2+a^2}\right)$$

$$\int \frac{dy}{\sqrt{x^2+y^2}} = \Big[\ln\left(y+\sqrt{y^2+x^2}\right)\Big]_{-a}^{+a} = \ln\Big[$$

x^2 → const

$$V(x) = \frac{kQ}{2a}\ln\left[\frac{\sqrt{a^2+x^2}+a}{\sqrt{a^2+x^2}-a}\right] \Rightarrow \text{unit } \text{an}$$

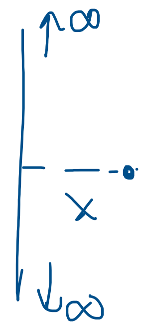


If $x\to\infty$ $\quad \frac{kQ}{x} = V(x)$

$\frac{2a}{x} \ll 1$

$$\sqrt{a^2+x^2}+a \longleftarrow \text{If } x \ll a$$

$$\sqrt{a^2\left(1+\frac{x^2}{a^2}\right)}+a \qquad 0 \simeq \frac{x}{a} \ll 1$$



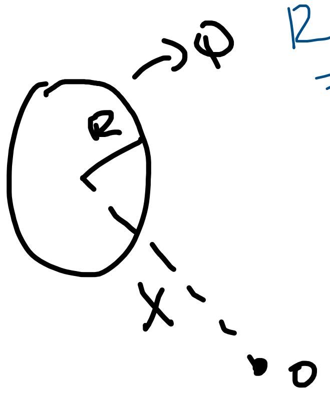
$V =$

$$a\left(1+\frac{x^2}{a^2}\right)^{1/2}+a$$

Taylor expansion $\Rightarrow \left[1+\left(\frac{x}{a}\right)^2\right.$

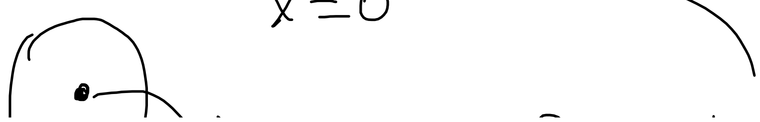
RING

$V_0 = ?$

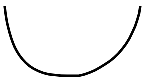


$\int \frac{k\,dq}{r}$

$x=0$



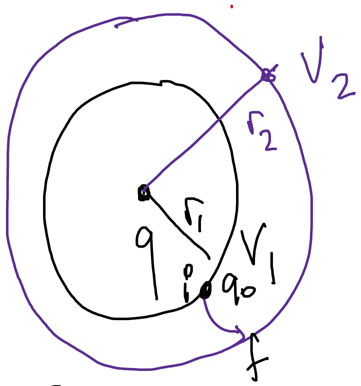
$V(x)$



$\rightarrow V(0) = \dfrac{kQ}{\sqrt{0^2+R^2}} = \underline{\underline{\dfrac{kQ}{R}}}$?

EQUIPOTENTIAL SURFACES

$\hookrightarrow$ (potential are equal)



$$V(r_1) = \frac{kq}{r_1} = V_1$$

$$V(r_2) = \frac{kq}{r_2} = V_2$$

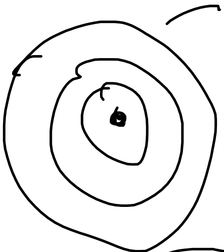
$\underline{\Delta U} = q\underline{\Delta V}$

$1 \to 2$

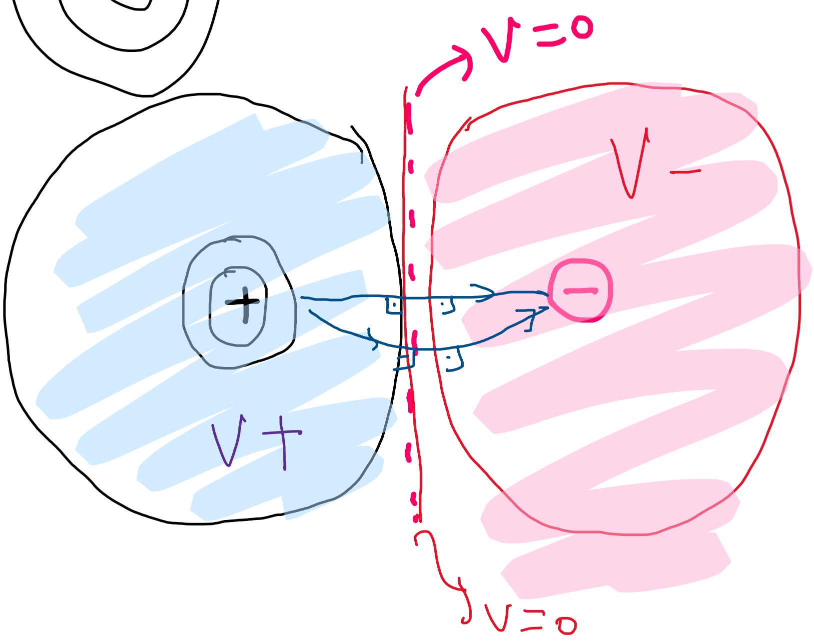
$\underline{U_2 - U_1} = q$

energy diff. =

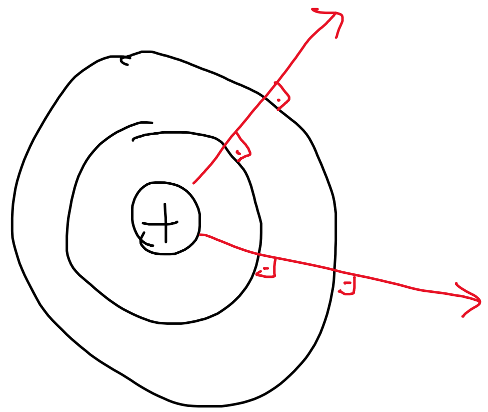
$\left\{ r_2 > r_1 \quad \Delta V < 0 \quad \Delta U < 0 \right\}$

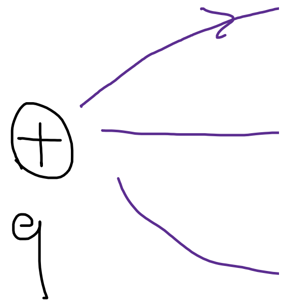


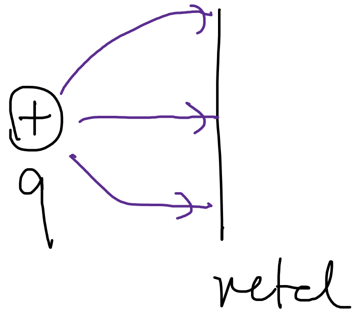
symmetric eq. pot. surfaces.



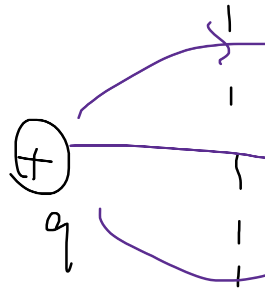
$\vec{E}$ field lines are perpendicular to Equipotential

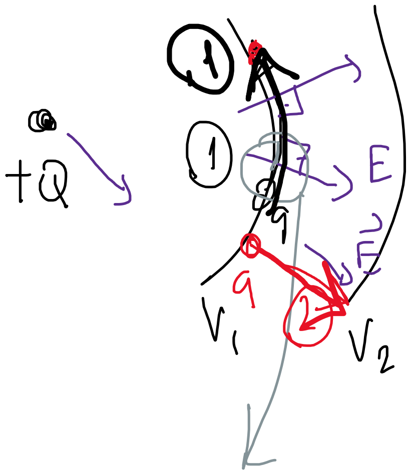






$\Leftrightarrow$





① q charge stays on V_1

$\Delta V = -\int \underbrace{\vec{E}\cdot d\vec{r}}$

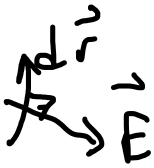
$V_1 - V_1 = 0$

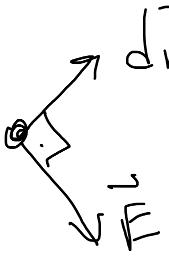
$E \neq 0 \quad dr$

$\vec{E}\cdot d\vec{r} = 0$

$E\,dr\,\underbrace{\cos\theta} = 0$

$\cos 90$





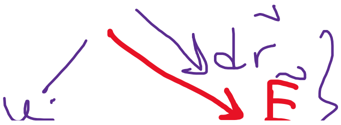
$\vec{E}\cdot d\vec{r} = 0 \}$ the reason $\vec{E} \perp$ Equip.

② path

$\Delta V = V_f - V_i = V_2 - V_1 = -\int \underbrace{\vec{E}\cdot d\vec{r}}_{\text{nonz}}$

$+$

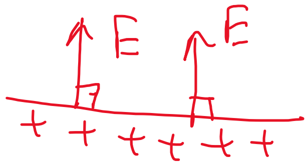
$V_2 - V_1 < 0$



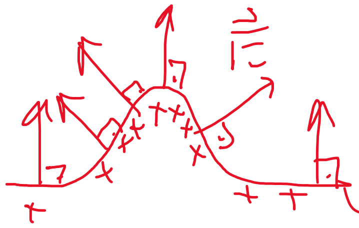
$V_2 < V_1$

V_p

metal surfaces at electrostatic equilibrium



→ metal surfaces are equipoten



still an equipotential s

if

electrostatic

$$\Delta V = -\int_i^f \vec{E}\cdot d\vec{r}$$

$$\int dV = -\int \vec{E}\cdot d\vec{r} \rightarrow dV = -\vec{E}\cdot d\vec{r}$$

$$= -\left(E_x\hat{i} + E_y\hat{j} + E_z\hat{k}\right)$$

$$E_x \neq -\frac{dV}{dx} \quad (\text{de})$$

$$\boxed{dV = -\left(E_x dx + E_y dy + \right.}$$

$$\begin{cases} E_x = -\dfrac{\partial V}{\partial x} \\ E_y = -\dfrac{\partial V}{\partial y} \qquad E_z = -\dfrac{\partial V}{\partial z} \end{cases}$$

$\partial \equiv$ partial derivative (del)

$V(x_1,$

ex $V = 10x^2y + 7xz + 4z^2$

$E_x = -(20$

$E_y = -($

E_z

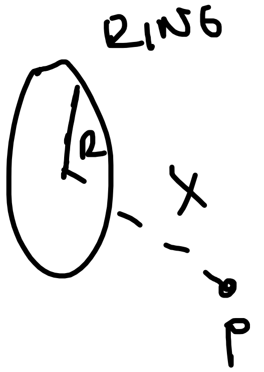
gradient

$\hookrightarrow \vec{\nabla} = \partial\,\hat{i} + \partial\,\hat{j} + \partial\,\hat{k}$

$\frac{\partial}{\partial x} \quad \frac{\partial}{\partial y} \quad \frac{\partial}{\partial z}$

$$\boxed{\vec{\nabla} V = \vec{E}}$$

RING



$$V = \frac{kQ}{\sqrt{R^2 + x^2}} \qquad \vec{E} = ?$$

$V(x) \Rightarrow \vec{E}$

$$-\frac{\partial V}{\partial x} = E_x = -kQ \underbrace{\frac{\partial (R^2 + x^2}{\partial x}}_{-\frac{1}{2}(R^2}$$

$$kQ \frac{\cancel{2}x}{\cancel{2}(x^2+R^2)^{3/2}} = E_x \checkmark$$

metal

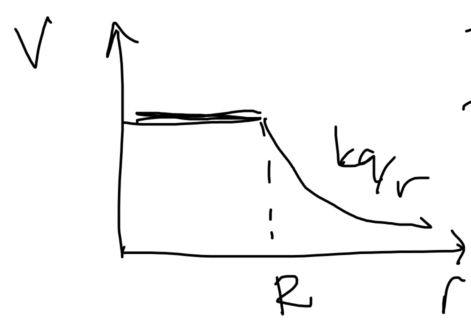
V=const

Inside V=const

$$V = \frac{kq}{r}$$

$$\vec{E}_r = -\frac{\partial}{\partial r}\left(\frac{kq}{r}\right)\hat{r} = -kq(-1)r^{-2} = \frac{kq}{r^2}$$

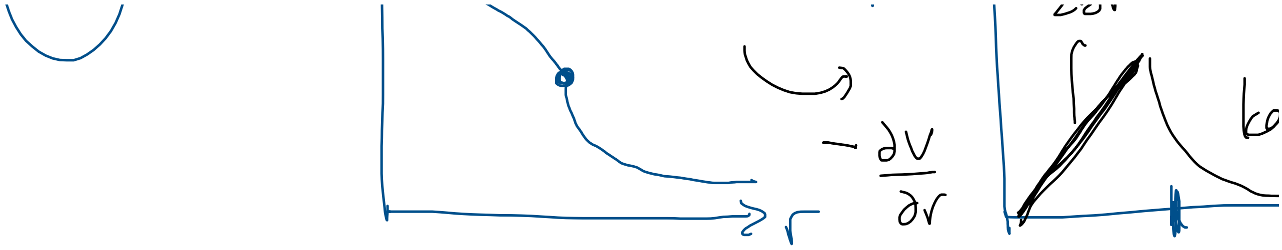
$$-\frac{\partial V}{\partial r} = 0 \Rightarrow \underbrace{\vec{E} = 0}$$



$-\frac{\partial V}{\partial r}$

Insulating / dielectric sphere

V $\quad A - Br^2 \qquad E \quad 2Br$



$$-\frac{\partial}{\partial r}\left(A - Br^2\right) = \underline{\underline{2Br}} = F(\text{inside})$$